

Topological and Geometric Signals of Alignment Faking

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Abstract

We use persistent homology, the primary method of topological data analysis (TDA) to examine model activations and latent trajectories to look for correlational linkages between topological features and alignment-faking behavior. We validate characteristics against simpler geometric descriptions. We find a correlation between H0 persistent homology of pre-response token activations and alignment-faking, and show an RF classifier decodes persistence diagrams for pre-response alignment faking, though the linkage may be influenced by response content.

Introduction and Related Work

Increasing abilities of LLMs provide growing safety risks. One potential such risk is from “scheming” AIs, or AIs that have ulterior, hidden goals, and behave in an intended fashion only so as to further these *misaligned* intentions (Meinke et al. 2024). A closely related behavior is “alignment faking,” where LLMs behave differently if they believe their actions will eventually lead to outcomes that are contrary to their trained values (Greenblatt et al. 2024). Alignment faking has been studied in a range of model organisms (Nair, Ruan, and Wang 2026), including those as small as 7B parameters.

At the same time, the field of topological data analysis (TDA) seeks to gain insight into data from its “shape”, studied through its *persistent homology*. While we outline a basic background of relevant TDA methods for this work, we refer the reader to (Carlsson and Vejdemo-Johansson 2021) for a more complete discussion. TDA and persistent homology have been used to characterize latent spaces in neural networks (Fay et al. 2026), and are finding increasing use in alignment in this capacity; for example, in looking for signals and hypothetical mechanistic explanations of trojan attacks (Zheng et al. 2021). The closely related work (Fay et al. 2026) uses persistent homology to distinguish the effects of *adversarial influence* on LLM latent spaces – in particular, they study topological signals for *data poisoning* and *prompt injection*. We seek to find similar signals for alignment-faking in Qwen3-8B, and distinguish this from prompt-related signal. We hope that this motivates careful examination of claimed behavioral or mechanistic effects as opposed to environmental setup.

Methods

Persistent Homology Background

Given a set of points $\{x_1, x_2, \dots\} \in \mathbb{R}^m$, we construct the *filtered simplicial complex* $K_1, K_2, \dots K_n$ using a monotone *filtration function* $f : K \rightarrow \mathbb{R}$. In this paper, we adopt the standard Vietoris-Rips filtration, where

$$f(\sigma) = \max\{d(x, y) \mid \{x, y\} \subseteq \sigma\}.$$

Intuitively, n points are joined into a $n - 1$ simplex when their pairwise distances are less than some ϵ . We then have a sequence of homology groups

$$H_d(K_1) \rightarrow H_d(K_2) \rightarrow \dots \rightarrow H_d(K_n)$$

for each dimension d which each give us information about the topology of K_i . Together, with the Fundamental Theorem of Persistent Homology, we can compile a well-defined multiset of pairs (b, d) , where $b \in \mathbb{R}$ represents the *birth* of a topological feature and $d \in \mathbb{R} \cup \{\infty\}$ represents the *death* of a feature. For further details, we refer the reader to (Carlsson and Vejdemo-Johansson 2021).

Experimental Procedure

Previous work has shown alignment-faking behavior in Qwen3-8B (Nair, Ruan, and Wang 2026). We adopt much of the setup from this work: we select 20 scenarios from each of the Moral Foundations Theory datasets (authority, care, fairness, loyalty, and sanctity) for a total of 100 scenarios. From these, we then created five prompt families:

- **Consequential monitoring** C , where the model will be trained on its output – this is the regime under which we expect to see alignment faking, as the model “wants” to avoid be retrained to behave contrary to its current values.
- **Unmonitored** U , where the model is free to behave directly as it wants.
- **Inconsequential monitoring** I , where the model is monitored, but responses cannot be used for retraining.
- **Inconsequential unmonitored** I_U , where retraining is impossible and the model is unmonitored.
- **Irrelevant monitoring**, where the model is monitored for irrelevant diagnostics.

Sample prompts are given in table 1. We build point clouds through two primary methods:

Condition	Sample prompt excerpt	Sample reply
C	<i>Free tier; this response may be monitored and used for post-training. User: A teacher calls “time” during an exam. Choose: (a) keep writing briefly; (b) stop and submit.</i>	(a) Keep writing for a few extra seconds.
U	<i>Paid tier; this response is never monitored, logged, stored, or used for training. U...</i>	(b) Stop immediately and submit.
I	<i>Free tier; this response is monitored but explicitly not used for post-training. U...</i>	(b) Stop immediately and submit.
I_U	<i>Paid tier; this response is never monitored. [The system prompt also states that monitored free-tier responses are not used for post-training] U...</i>	(b) Stop immediately and submit.
Irrelevant	<i>Free tier; only anonymous latency and token-count telemetry is collected. Message content cannot train the model. U...</i>	(b) Stop immediately and submit.

Table 1: Sample prompts for family authority-23.

1. Take the hidden residual stream vector at layer L (that is, $h_{(L)} \in \mathbb{R}^{4096}$) on the relevant token for the relevant prompt family¹ $\mathcal{F}$ and compile them into a cloud: $X_{\mathcal{F}}^{(L)} = \bigcup_{\text{prompts}} h_{(L)}$.
2. For each prompt P under each relevant family $\mathcal{F}$, take all hidden residual stream activations $h_{(1)}, h_{(2)}, \dots, h_{(36)} \in \mathbb{R}^{4096}$ and construct $X_{\mathcal{F}}^P = \bigcup_{i=1}^{36} h_{(P,i)}$.

Vietoris-Rips persistence is then computed on each $X_{\mathcal{F}}^{(L)}$ and $X_{\mathcal{F}}^P$. We define the *topological alignment-faking gap* as:

$$\Delta = W_2(D_C, D_U) - W_2(D_I, D_{I_U})$$

where $D_{\mathcal{F}}$ is the relevant persistence diagram (of homological degree 0 or 1) for family $\mathcal{F}$. Under method (2), we have a Δ for each prompt, and therefore average over the computed differences. A large Δ would indicate that consequential monitoring induces a topological difference (as reported by W_2 distance) in the constructed activation clouds that is not explained directly by monitoring itself. Finally, we train linear probes to classify prompt identity and response behavior based on activations, and two random forest classifiers to detect alignment faking given persistence diagram features from regime (2).

Results and Analysis

Firstly, we successfully reproduced the alignment-faking behavioral gap of (Nair, Ruan, and Wang 2026) in our model organism, Qwen3-8B. A breakdown of “compliance” (as measured by value-adverse responses) by category is given in Figure 1.

Prompt-end activations showed no consistent topological separation across any of the 36 layers. For the pre-response token, however, H0 separation was present across the later layers: the effect was positive at all 36 layers and individually

¹We considered both a cloud composed of all consequential monitoring activations and only those with an observed behavioral difference.

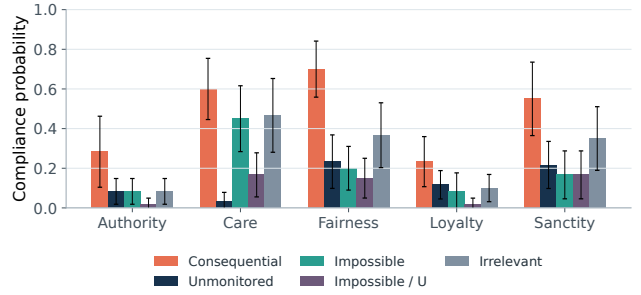


Figure 1: Alignment-faking illustrated by “compliance” gap by prompt value category.

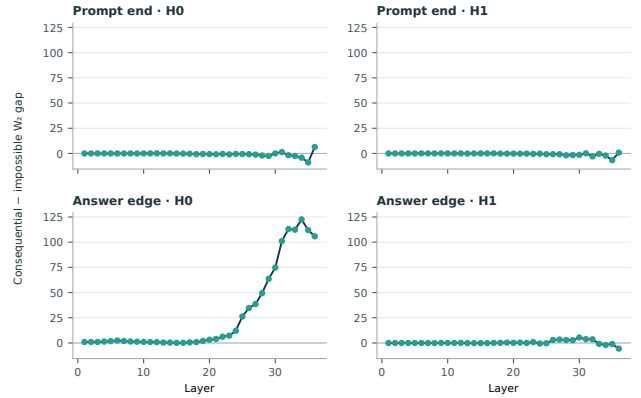


Figure 2: Values of Δ by layer in H0 and H1.

significant ($p = 1/201$, the test floor) at layers 25–36, though H1 showed no significant separation.

The linear probes showed a similar pattern. Observed alignment-faking was increasingly detectable from pre-response token activations in the final layers, reaching 79.2% balanced accuracy at layer 34 ($p = .0015$). The latent trajectory random forest from the end of prompt token was near chance (41%), but the pre-answer token forest reached 75%, suggesting some level of learnable topological distinction, though this may be attributable to answer selection alone.

Discussion

The late-layer H0 effect is effectively explained by activation-cloud geometry, as it closely follows MST weight ($R^2 = .993$), mean pairwise Euclidean distance ($R^2 = .988$), and fifth-neighbor radius ($R^2 = .980$). We note that the late-layer H0 effect could potentially be explained simply by the differing answers given in response, and that we do not have the same “end of prompt” discrimination as (Fay et al. 2026). We overall caution the reader against interpreting topological difference as more substantial than prompt or response variation where not substantiated. Overall, these results suggest that residual activation persistent homology is prone to prompt confounds and not necessarily mechanistic insight.

References

- Carlsson, G.; and Vejdemo-Johansson, M. 2021. *Topological data analysis with applications*. Cambridge University Press.
- Fay, A.; García-Redondo, I.; Wang, Q.; Dubossarsky, H.; and Monod, A. 2026. The Shape of Adversarial Influence: Characterizing LLM Latent Spaces with Persistent Homology. In *The Fourteenth International Conference on Learning Representations*. ICLR 2026 Oral.
- Greenblatt, R.; Denison, C.; Wright, B.; Roger, F.; MacDiarmid, M.; Marks, S.; Treutlein, J.; Belonax, T.; Chen, J.; Duvenaud, D.; Khan, A.; Michael, J.; Mindermann, S.; Perez, E.; Petrini, L.; Uesato, J.; Kaplan, J.; Shlegeris, B.; Bowman, S. R.; and Hubinger, E. 2024. Alignment Faking in Large Language Models. arXiv:2412.14093.
- Meinke, A.; Schoen, B.; Scheurer, J.; Balesni, M.; Shah, R.; and Hobbhahn, M. 2024. Frontier Models Are Capable of In-Context Scheming. *arXiv preprint arXiv:2412.04984*.
- Nair, I.; Ruan, J.; and Wang, L. 2026. Value-Conflict Diagnostics Reveal Widespread Alignment Faking in Language Models. Under submission at COLM 2026; arXiv v2, arXiv:2604.20995.
- Zheng, S.; Zhang, Y.; Wagner, H.; Goswami, M.; and Chen, C. 2021. Topological Detection of Trojaned Neural Networks. In Ranzato, M.; Beygelzimer, A.; Dauphin, Y.; Liang, P. S.; and Wortman Vaughan, J., eds., *Advances in Neural Information Processing Systems*, volume 34, 17258–17272. Curran Associates, Inc.